

Linear Algebra — Week 01

Fields, Vector Space Axioms & Complex Numbers

Why this matters

Algebra is about **definitions, theorems, proofs, and structure**. Every structure we study is built from a small list of **axioms** — basic rules used to describe how mathematical objects behave.

DEFINITION Binary Operation

A **binary operation** is a function $\boxplus : X \times X \rightarrow X$ with two inputs and one output.

- Real addition $+$: $\mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is a binary operation.
- Real multiplication $\cdot$: $\mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is a binary operation.

DEFINITION Commutative and Associative

An operation $\boxplus : X \times X \rightarrow X$ is **commutative** if, for all $x, y \in X$:

$$x \boxplus y = y \boxplus x.$$

An operation $\boxplus : X \times X \rightarrow X$ is **associative** if, for all $x, y, z \in X$:

$$x \boxplus (y \boxplus z) = (x \boxplus y) \boxplus z.$$

Non-commutative operators: division $\left(\frac{2}{3} \neq \frac{3}{2}\right)$, subtraction $(2 - 3 \neq 3 - 2)$, exponents $(2^3 \neq 3^2)$.

Non-associative operators: subtraction $((1 - 2) - 3 \neq 1 - (2 - 3))$, exponents $((2^3)^4 \neq 2^{(3^4)})$.

DEFINITION The Axioms of a Field

A **field** is a set F together with two binary operations

$$\boxplus : F \times F \rightarrow F, \quad \boxtimes : F \times F \rightarrow F$$

called *addition* and *multiplication*, such that:

- **A1.** $\boxplus$ is associative.
- **A2.** $\boxplus$ is commutative.
- **A3.** There exists an additive identity $\hat{0} \in F$ and a multiplicative identity $1 \in F$ such that $\hat{0} \boxplus x = x$ and $1 \boxtimes x = x$ for all $x \in F$.
- **A4.** For each $x \in F$ there is an additive inverse $x' \in F$ such that $x \boxplus x' = \hat{0}$.
- **A5.** $\boxtimes$ is associative.
- **A6.** $\boxtimes$ is commutative.
- **A7.** For each $x \in F$ except $\hat{0}$ there is a multiplicative inverse $x' \in F$ such that $x \boxtimes x' = 1$.
- **A8.** Multiplication is distributive over addition: for all $a, b, c \in F$,
$$a \boxtimes (b \boxplus c) = a \boxtimes b \boxplus a \boxtimes c.$$

Their order, numbering, and names are important. This course uses two fields: the **real numbers** $\mathbb{R}$ and the **complex numbers** $\mathbb{C}$.

DEFINITION The Imaginary Number i

$$i = \sqrt{-1}.$$

i is not a real number, but it plays a central role in mathematics.

DEFINITION Complex Numbers

A complex number has the form

$$z = a + bi, \quad a, b \in \mathbb{R},$$

where $a = \operatorname{Re}(z)$ is the *real part* and $b = \operatorname{Im}(z)$ is the *imaginary part*.

DEFINITION Complex Addition and Multiplication

Let $z_1 = a + bi$ and $z_2 = c + di$. Then

$$z_1 + z_2 := (a + c) + (b + d)i, \quad z_1 \times z_2 := (ac - bd) + (ad + bc)i,$$

where the operations inside the brackets are the standard real-number operations.

DEFINITION Additive Inverse, Subtraction, Conjugate, Modulus, Division

- Additive inverse: $-z := (-a) + (-b)i$.
- Subtraction: $z_1 - z_2 := (a - c) + (b - d)i$.
- Multiplicative inverse (a, b not both 0):

$$z_1^{-1} := \left(\frac{a}{a^2 + b^2} \right) + \left(\frac{-b}{a^2 + b^2} \right) i.$$

- Complex conjugate: $\bar{z} := a - bi$.
- Modulus: $|z| = \sqrt{\operatorname{Re}(z)^2 + \operatorname{Im}(z)^2}$.
- Division: $\frac{z_1}{z_2} := z_1 \times (z_2)^{-1} = \frac{z_1 \bar{z}_2}{|z_2|^2}$.

DEFINITION The Axioms of a Vector Space

A **vector space** is a set V together with a field F and two binary operations

$$\boxplus : V \times V \rightarrow V, \quad \boxdot : F \times V \rightarrow V$$

called *vector addition* and *scalar multiplication*, such that:

- **A1.** $\boxplus$ is associative.
- **A2.** $\boxplus$ is commutative.
- **A3.** There exists $\vec{0} \in V$ (the **zero vector**) such that $\vec{0} \boxplus x = x$ for all $x \in V$.
- **A4.** For each $x \in V$ there is an additive inverse $x' \in V$ such that $x \boxplus x' = \vec{0}$.
- **A5.** For all $x, y \in V$ and $c \in F$: $c \boxdot (x \boxplus y) = (c \boxdot x) \boxplus (c \boxdot y)$.
- **A6.** For all $x \in V$ and $c, d \in F$: $(c + d) \boxdot x = (c \boxdot x) \boxplus (d \boxdot x)$.
- **A7.** For all $x \in V$ and $c, d \in F$: $(cd) \boxdot x = c \boxdot (d \boxdot x)$.
- **A8.** For all $x \in V$: $1 \boxdot x = x$.

(Axioms adapted from Little and Damiano, p. 7.)

A **complex vector space** ($\mathbb{C}$ -vector space) is a vector space over the field $\mathbb{C}$. A **real vector space** ($\mathbb{R}$ -vector space) is a vector space over the field $\mathbb{R}$.

KEY FACT Proof Techniques Introduced This Week

- **Direct proof:** prove a statement true or false using only established definitions, axioms, lemmas, or theorems — no extra assumptions.
- **Uniqueness proof:** assume two objects both have the desired property, then show they must be equal to each other.

Worked Examples

EXAMPLE 1

Verify the additive inverse of a complex number

Let $z = a + bi$. Show $z + (-z) = 0$ for all $z \in \mathbb{C}$.

$$z + (-z) = (a + bi) + (-a + (-b)i) = (a + (-a)) + (b + (-b))i = 0 + 0i = 0. \quad \blacksquare$$

EXAMPLE 2

Verify the multiplicative inverse of a complex number

Let $z_1 = a + bi$, with a, b not both 0. Show $z_1 \times z_1^{-1} = 1$.

$$(a + bi) \left(\left(\frac{a}{a^2 + b^2} \right) + \left(\frac{-b}{a^2 + b^2} \right) i \right) = \left(\frac{a^2 + b^2}{a^2 + b^2} \right) + \left(\frac{-ab + ab}{a^2 + b^2} \right) i = 1 + 0i. \quad \blacksquare$$

EXAMPLE 3

Uniqueness of the zero vector

Show the zero vector $\vec{0} \in V$ (Axiom A3) is unique.

Proof. Suppose $\vec{0}$ and $\vec{0}'$ both satisfy A3. Then

$$\vec{0}' = \vec{0} \boxplus \vec{0}' \quad (\text{A3 for } \vec{0}, \text{ with } x = \vec{0}')$$
$$= \vec{0}' \boxplus \vec{0} \quad (\text{A2, commutativity}) = \vec{0} \quad (\text{A3 for } \vec{0}', \text{ with } x = \vec{0}).$$

So $\vec{0} = \vec{0}'$. $\blacksquare$

EXAMPLE 4

A fact that seems obvious but needs proof

If $\vec{z} \in V$, then $0\vec{z} = \vec{0}$.

Proof.

$$0\vec{z} = (0 + 0)\vec{z} = 0\vec{z} \boxplus 0\vec{z} \quad (\text{A6}).$$

Add the additive inverse $(0\vec{z})'$ to both sides:

$$0\vec{z} \boxplus (0\vec{z})' = (0\vec{z} \boxplus 0\vec{z}) \boxplus (0\vec{z})' \Rightarrow \vec{0} = 0\vec{z} \boxplus (0\vec{z} \boxplus (0\vec{z})') \quad (\text{A1, associativity})$$
$$= 0\vec{z} \boxplus \vec{0} = \vec{0} \boxplus 0\vec{z} \quad (\text{A2}) = 0\vec{z} \quad (\text{A3}).$$

Hence $0\vec{z} = \vec{0}$ in **any** vector space. $\blacksquare$

EXAMPLE 5

$(-1)x$ is the additive inverse of x

Let $x \in V$. Prove $(-1)x$ is an additive inverse of x .

$$x \boxplus (-1)x = 1x \boxplus (-1)x \quad (\text{A8}) = (1 + (-1))x \quad (\text{A6}) = 0x = \vec{0} \quad (\text{Example 4}). \quad \blacksquare$$

Practice Problems

Read the hint and the caution before you start each problem — not after.

Complex numbers

1. Let $z_1 = 3 - 2i$ and $z_2 = -1 + 4i$. Compute $z_1 + z_2$, $z_1 - z_2$, and $z_1 \times z_2$.

HINT Combine real and imaginary parts separately for the sum/difference, and expand the product term by term using $i^2 = -1$.

CAUTION — before you solve Watch out: when expanding the product, don't leave the $bd i^2$ term unsimplified — remember $i^2 = -1$.

Your working:

2. Compute the modulus $|z|$ for $z = 5 - 12i$.

HINT Apply the modulus formula $|z| = \sqrt{\text{Re}(z)^2 + \text{Im}(z)^2}$ using both components of z .

CAUTION — before you solve Don't confuse the modulus with just $|\text{Re}(z)|$ — you must include the imaginary part too.

Your working:

3. Find z^{-1} for $z = 2 + 3i$, then verify $z \times z^{-1} = 1$.

HINT Use the multiplicative-inverse formula from Page 1 directly — don't try to guess-and-check.

CAUTION — before you solve Watch out for a sign error: the formula uses $-\frac{b}{a^2 + b^2}$, not $+\frac{b}{a^2 + b^2}$.

Your working:

4. Compute $\frac{2+i}{1-i}$ by rationalizing with the conjugate.

HINT Multiply both the numerator and denominator by the conjugate of the denominator before simplifying.

CAUTION — before you solve Remember: you must multiply *both* the numerator and the denominator by the conjugate, not just one.

Your working:

5. Find the complex conjugate of $z = -4 + 7i$ and confirm $z + \bar{z}$ is a real number.

HINT Recall exactly which part of z the conjugate operation changes the sign of.

CAUTION — before you solve Don't assume the conjugate flips the sign of the real part a as well as b — only b changes sign.

Your working:

Fields

6. State axioms A1–A8 for a field in your own words (without looking back at Page 1), then check your answer against the definition.

HINT Go through A1–A8 one at a time, naming the single property (associativity, commutativity, identity, etc.) each one governs.

CAUTION — before you solve Watch out: don't mix up A3 (identity) with A4 (inverse), and remember A3 requires *both* an additive and a multiplicative identity.

Your working:

7. Explain, using an example, why subtraction is **not** a binary operation that makes $\mathbb{R}$ satisfy Axiom A2 (commutativity) on its own — i.e. why subtraction is not the “addition” operation of a field.

HINT Test commutativity by trying a concrete pair of numbers in the operation, not just checking the operation is defined.

CAUTION — before you solve Don't settle for showing subtraction is defined for one pair like $2 - 3$ — the axiom demands commutativity for *all* pairs.

Your working:

8. Is $(\mathbb{Z}, +, \times)$ (the integers with standard addition and multiplication) a field? Identify which axiom fails and explain why.

HINT Check each field axiom systematically for $(\mathbb{Z}, +, \times)$, paying special attention to inverses.

CAUTION — before you solve Don't assume closure under $+$ and $\times$ is enough — remember inverses must also lie inside the set.

Your working:

Vector space axioms

9. Let $x, y \in V$ and suppose $x \boxplus y = x$. Prove that $y = \vec{0}$. (Hint: add the additive inverse of x to both sides.)

HINT Add the additive inverse of x to both sides of $x \boxplus y = x$ and work through the axioms explicitly.

CAUTION — before you solve Don't informally "cancel" x from both sides — add the inverse and cite A1/A2/A3/A4 explicitly.

Your working:

10. Prove that the additive inverse of any $x \in V$ is unique. (Model your proof on the zero-vector uniqueness proof on Page 3.)

HINT Model your proof on the zero-vector uniqueness proof, assuming two inverses and showing they're equal.

CAUTION — before you solve Remember: don't skip using commutativity (A2) — it's the key pivot step in the argument.

Your working:

11. Let $x \in V$. Using the axioms only, prove that $-(-x) = x$.

HINT Rewrite $-x$ as $(-1)x$ using Example 5, then build the proof from the axioms rather than intuition.

CAUTION — before you solve Don't treat $-x$ as some new independent object — route through $(-1)x$ from Example 5 instead.

Your working:

12. A student claims: “Since $1 \cdot x = x$ for all x (A8), it must also be true that $2 \cdot x = x \oplus x$ for all x .” Prove this claim using the axioms (do not just assert it).

HINT Write 2 as $1 + 1$ and apply Axiom A6 before invoking A8.

CAUTION — before you solve Don’t assume $2x$ “obviously” equals $x \oplus x$ — you must justify it from the axioms, not assert it.

Your working:
